PROJECT REPORT

Customer Behavior Analytics

An end-to-end analysis of 3,900 customers using Python, SQL and Power BI

Yuthika · October 2026

THE BUSINESS AT A GLANCE

3,900 customers spend an average of $59.76 per purchase

3,900	$233,081	$59.76	3.75 / 5
Customers	Total revenue	Average purchase	Average review rating

The data covers clothing, accessories, footwear and outerwear purchases, with 18 attributes per customer.

WORKFLOW

Four tools take the raw file to a finished dashboard

1

Prepare

Python (pandas)

Clean the data and engineer new features

2

Store

MySQL

Load the cleaned table with SQLAlchemy

3

Analyse

SQL

Ten business questions using CTEs, window functions and subqueries

4

Visualise

Power BI

An interactive dashboard with filters

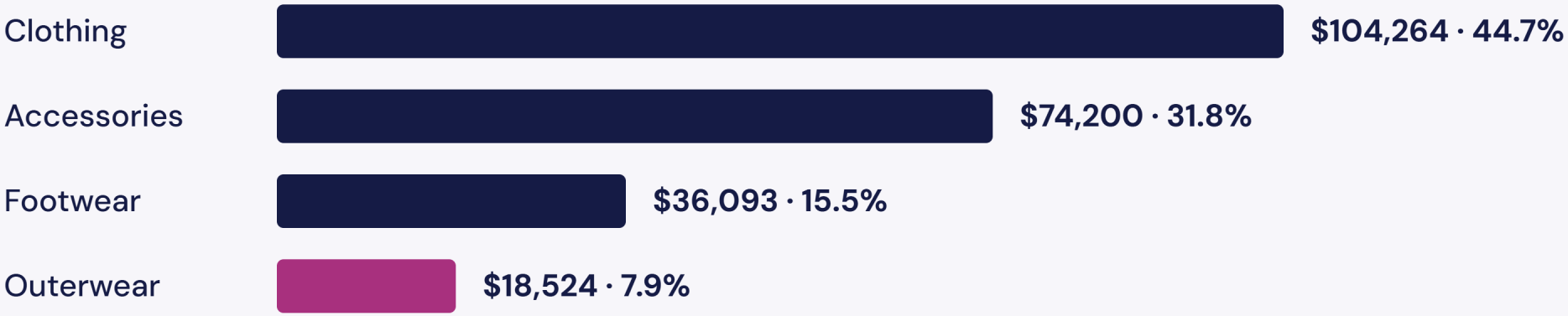
DATA PREPARATION

Five cleaning steps made the data analysis-ready

Step	What was done
Missing ratings	37 values filled with the category median
Column names	Lower-case with underscores, ready for SQL
Age groups	Four quartile bins: 18–31, 32–44, 45–57, 58–70
Redundant column	promo_code_used matched discount_applied, so it was dropped
Database load	3,900 rows × 19 columns written to MySQL

REVENUE BY CATEGORY

Clothing and Accessories generate 76.5% of revenue



Average purchase is $57–$60 in every category, so how often items sell drives the gap, not price.

GENDER

Men bring in 67.7% of revenue because they are 68% of customers

Metric	Male	Female
Total revenue	$157,890	$75,191
Customers	2,652	1,248
Average per customer	$59.54	$60.25

Spend per customer is almost identical, so the opportunity is acquiring more female customers.

SUBSCRIPTIONS

Subscribers spend about the same as non-subscribers

27%

of customers are subscribed (1,053 of 3,900)

$59.49

average spend of subscribers, against $59.87 for non-subscribers

27.6%

of repeat buyers subscribe, close to the 27.0% overall rate

Loyalty and subscription are not linked, which leaves 2,518 loyal non-subscribers as a ready target group.

DISCOUNTS

43% of purchases use a discount, and half of those customers still spend above average

43%

of all purchases carry a discount

50%

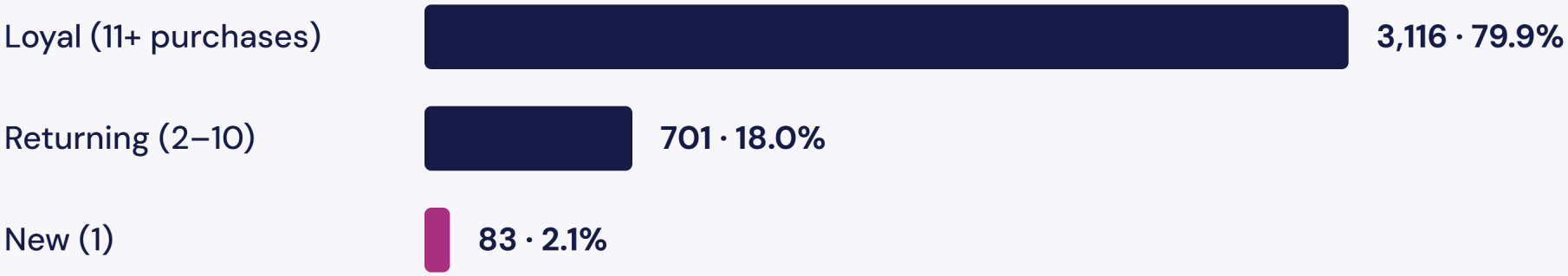
of discount users spent at least the $59.76 average (839 of 1,677)

Most-discounted products

Product	Discount rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

CUSTOMER LOYALTY

80% of customers are loyal, but only 2% are new



Retention is strong; bringing in new customers is the weak spot.

AGE AND SHIPPING

Spend is balanced across age groups and shipping types

Revenue by age group

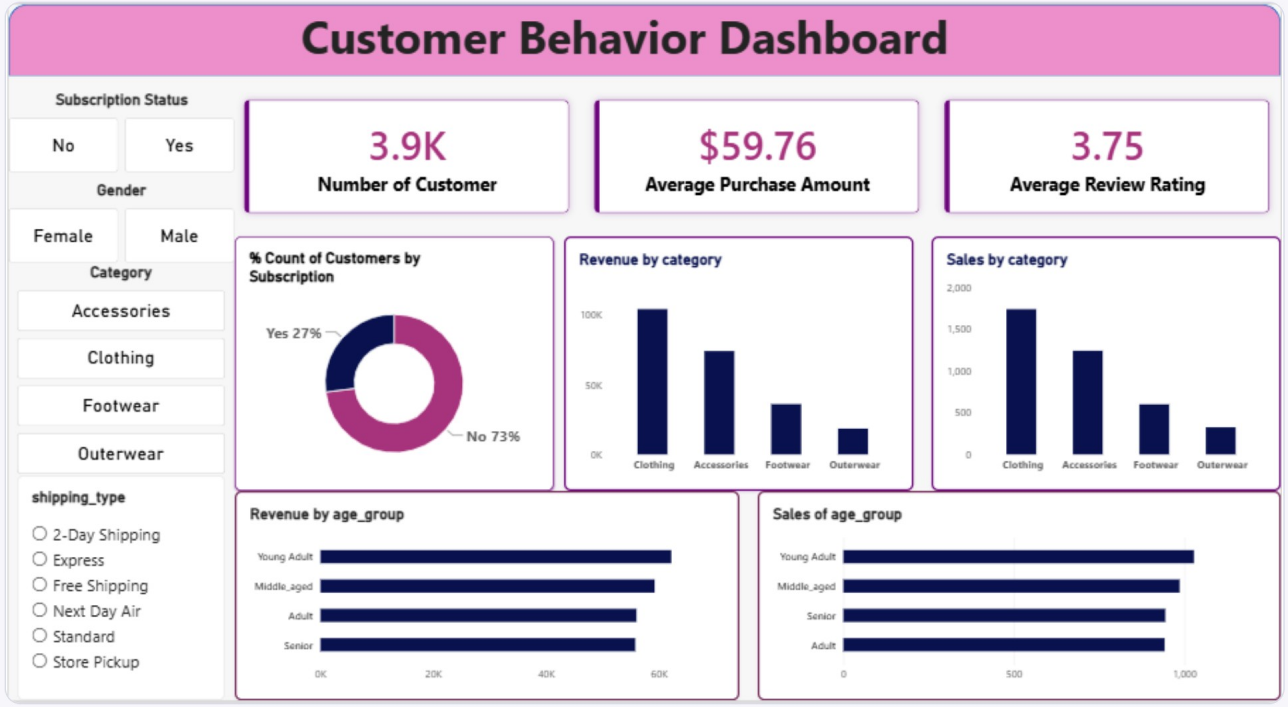
Age group	Revenue	Share
Young Adult (18–31)	$62,143	26.7%
Middle-aged (45–57)	$59,197	25.4%
Adult (32–44)	$55,978	24.0%
Senior (58–70)	$55,763	23.9%

Average purchase by shipping

Shipping type	Average
2-Day Shipping	$60.73
Express	$60.48
Free Shipping	$60.41
Store Pickup	$59.89
Next Day Air	$58.63
Standard	$58.46

POWER BI DASHBOARD

One interactive view answers the common questions



- KPI cards for customers, average purchase and rating
- Slicers for subscription, gender, category and shipping
- Revenue and sales by category and by age group

RECOMMENDATIONS

Four actions follow from the analysis

Redesign the subscription

Add benefits that change behavior, starting with the 2,518 loyal non-subscribers.

Target discounts

Half of discount users already spend above average, so review the five most-discounted products first.

Acquire new customers

Only 2.1% are new. Focus on female customers, who spend as much per head but are 32% of the base.

Grow Outerwear

Outerwear is 7.9% of revenue. Bundle it with Clothing purchases and seasonal campaigns.

TAKEAWAYS AND NEXT STEPS

A loyal, evenly spending customer base with clear room to grow

Add dates

Purchase dates would unlock trend and seasonality analysis.

Segment with RFM

Replace fixed thresholds with data-driven customer segments.

Predict subscription

Build a classification model to find likely subscribers.

Limits: one purchase per customer and no dates, so findings show association, not cause.